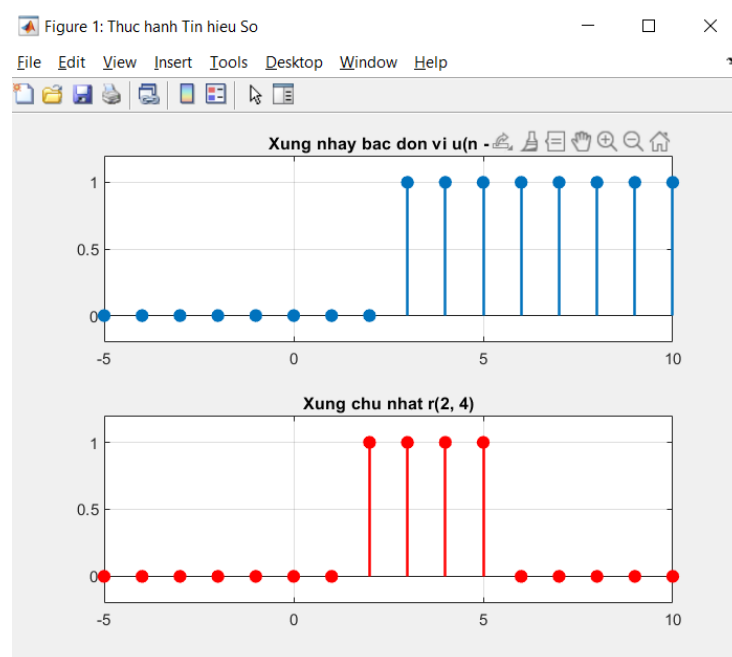


Bài 1:

Code

Scrip1	Scrip2	Scrip3
<pre>clear; clc; close all; n1 = -5; n2 = 10; [u, n_u] = unit_step(3, n1, n2); [r, n_r] = rect_sig(2, 4, n1, n2); figure('Name', 'Thuc hanh Tin hieu So'); subplot(2,1,1); stem(n_u, u, 'filled', 'LineWidth', 1.5); title('Xung nhay bac don vi u(n - 3)'); grid on; axis([n1 n2 -0.2 1.2]); subplot(2,1,2); stem(n_r, r, 'filled', 'r', 'LineWidth', 1.5); title('Xung chu nhac r(2, 4)'); grid on; axis([n1 n2 -0.2 1.2]);</pre>	<pre>function [x, n] = unit_step(n0, n1, n2) n = n1:n2; x = (n >= n0); end</pre>	<pre>function [x, n] = rect_sig(n0, L, n1, n2) n = n1:n2; x = (n >= n0) & (n < n0 + L); end</pre>

Kết quả chạy



Bài 2:

Code

```
clear; clc; close all;

t1 = -20:0.01:20;
y1 = sin(t1)./t1;
y1(t1==0) = 1;

t2 = 0:0.01:5;
y2 = exp(-t2);

T = 2;
t3 = -10:0.01:10;
y3 = (square(2*pi/T * t3, 50) + 1) / 2;

t4 = -50:0.01:50;
y4 = sin(pi*t4/3 + pi/4) + 2*cos(pi*t4/4);

figure('Name', 'Bai tap 2: Tin hieu lien tuc');

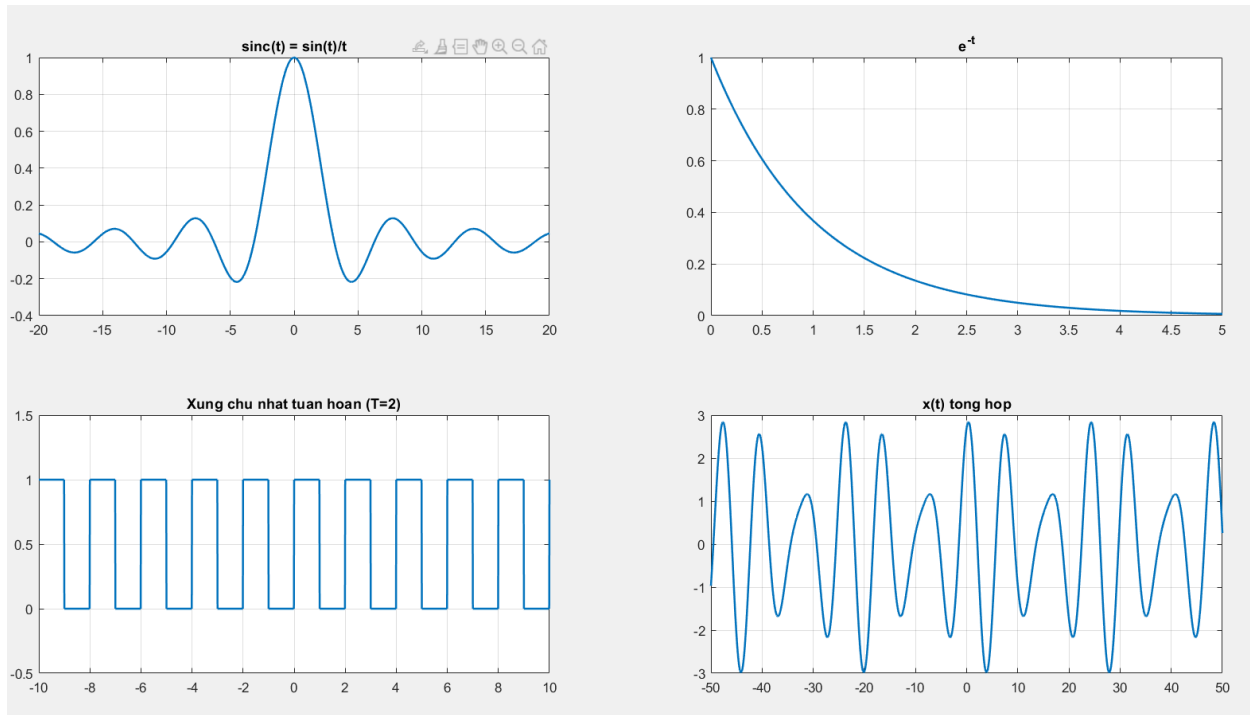
subplot(2,2,1);
plot(t1, y1, 'LineWidth', 1.5);
title('sinc(t) = sin(t)/t'); grid on;

subplot(2,2,2);
plot(t2, y2, 'LineWidth', 1.5);
title('e^{-t}'); grid on;

subplot(2,2,3);
plot(t3, y3, 'LineWidth', 1.5);
axis([-10 10 -0.5 1.5]);
title('Xung chu nhat tuan hoan (T=2)'); grid on;

subplot(2,2,4);
plot(t4, y4, 'LineWidth', 1.5);
title('x(t)'); grid on;
```

Kết quả



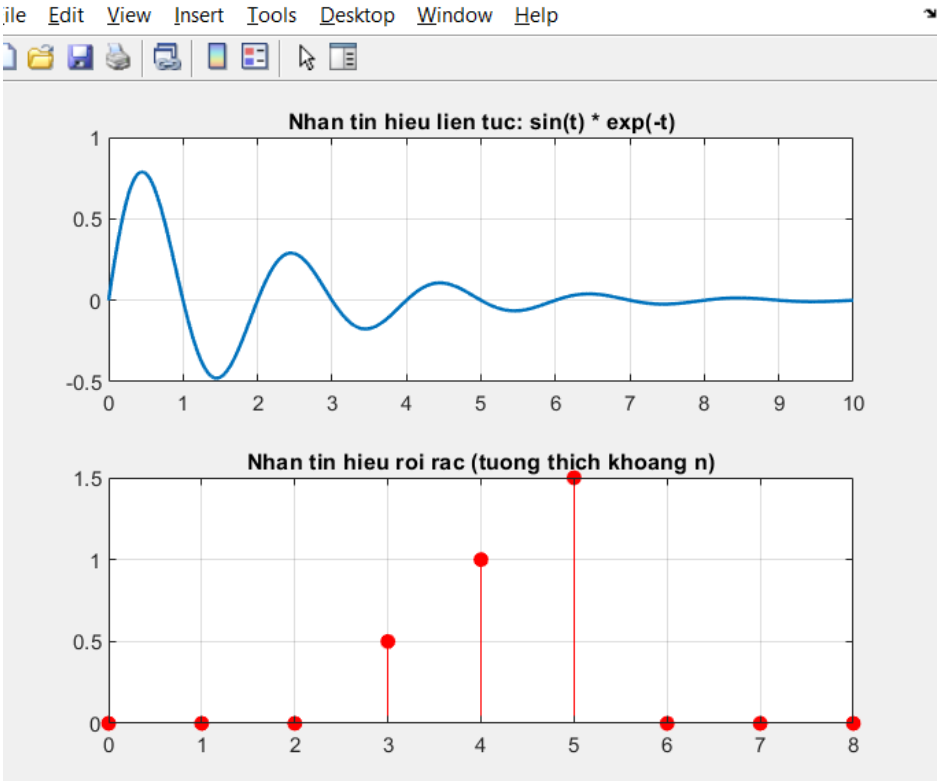
Bài 3

Code

Scrip			
<pre>function [y, n] = sig_mult(x1, n1, x2, n2) n = min(min(n1), min(n2)) : max(max(n1), max(n2)); y1 = zeros(1, length(n)); y2 = zeros(1, length(n)); y1((n >= min(n1)) & (n <= max(n1))) = x1; y2((n >= min(n2)) & (n <= max(n2))) = x2; y = y1 .* y2; end</pre>	<pre>function [y, t] = sig_mult_cont(x1, x2, t) y = x1 .* x2; end</pre>	<pre>function y = sig_scale(x, a) y = a * x; end function y = sig_add_const(x, a) y = x + a; end</pre>	<pre>clear; clc; close all; t = 0:0.01:10; x1_c = sin(2*pi*0.5*t); x2_c = exp(-0.5*t); y_c = x1_c .* x2_c; n1 = 0:5; x1_d = [1 1 1 1 1 1]; n2 = 3:8; x2_d = [0.5 1 1.5 2 2.5 3]; [y_d, n_d] = sig_mult(x1_d, n1, x2_d, n2); figure; subplot(2,1,1); plot(t, y_c, 'LineWidth', 1.5);</pre>

			<pre> title('Nhan tin hieu lien tuc: sin(t) * exp(- t)'); grid on; subplot(2,1,2); stem(n_d, y_d, 'filled', 'r'); title('Nhan tin hieu roi rac (tuong thich khoang n)'); grid on; </pre>
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Kết quả



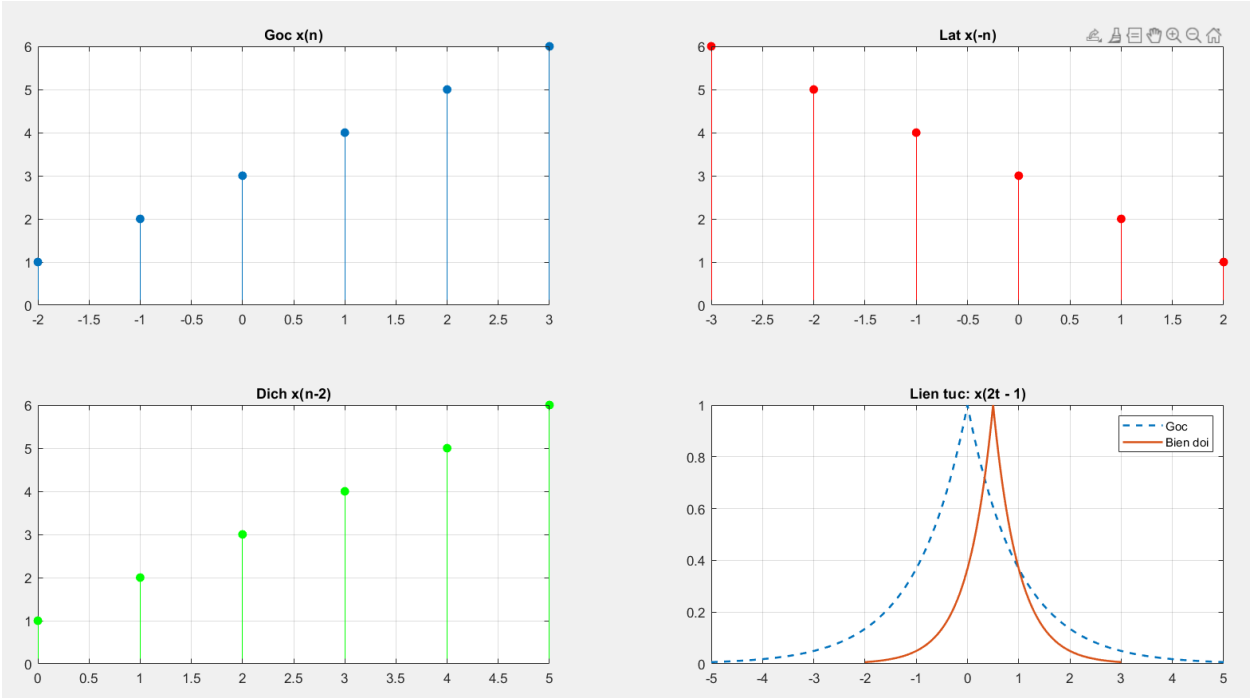
Bài 4

Code

<pre> clear; clc; close all; n = -2:3; x = [1 2 3 4 5 6]; </pre>	<pre> function [y, t_new] = transform_cont(x, t, a, b) </pre>	<pre> function [y, n_new] = sig_shift(x, n, k) </pre>	<pre> function [y, n_new] = sig_fold(x, n) </pre>	<pre> function [y, n_new] = sig_downsample(x, n, M) </pre>
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<pre> [x_f, n_f] = sig_fold(x, n); [x_s, n_s] = sig_shift(x, n, 2); t = -5:0.01:5; x_c = exp(-abs(t)); [y_c, t_c] = transform_cont(x_c, t, 2, -1); figure; subplot(2,2,1); stem(n, x, 'filled'); title('Goc x(n)'); grid on; subplot(2,2,2); stem(n_f, x_f, 'filled', 'r'); title('Lat x(-n)'); grid on; subplot(2,2,3); stem(n_s, x_s, 'filled', 'g'); title('Dich x(n-2)'); grid on; subplot(2,2,4); plot(t, x_c, '--', t_c, y_c, 'LineWidth', 1.5); legend('Goc', 'Bien doi'); title('Lien tuc: x(2t - 1)'); grid on; </pre>	<pre> t_new = (t - b) / a; y = x; end </pre>	<pre> % y(n) = x(n - k) n_new = n + k; y = x; end </pre>	<pre> y = fliplr(x); n_new = -fliplr(n); end </pre>	<pre> y = x(1:M:end); n_new = n(1:M:end) / M; end </pre>
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Kết quả



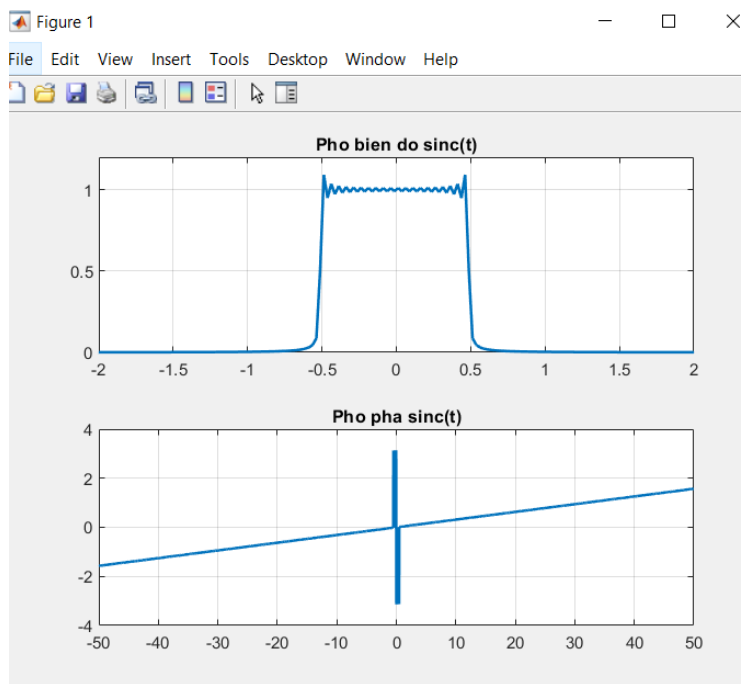
Bài 5

Code

Câu a	Câu b
<pre>Fs = 100; T = 20; t = -T/1/Fs:T; x = sinc(t); L = length(x); X = fftshift(fft(x)); f = (-L/2:L/2-1)*(Fs/L); figure; subplot(2,1,1); plot(f, abs(X)/Fs, 'LineWidth', 1.5); title('Pho bien do sinc(t)'); grid on; axis([-2 2 0 1.2]); subplot(2,1,2); plot(f, angle(X), 'LineWidth', 1.5); title('Pho pha sinc(t)'); grid on;</pre>	<pre>Fs = 10; T = 100; t = 0:1/Fs:T; x = cos(pi*t/3) + 2*sin(pi*t/4); L = length(x); X = fftshift(fft(x)); f = (-L/2:L/2-1)*(Fs/L); figure; subplot(2,1,1); plot(f, abs(X)/(L/2), 'LineWidth', 1.5); title('Pho bien do x(t)'); grid on; axis([-1 1 0 2.2]); subplot(2,1,2); plot(f, angle(X), 'LineWidth', 1.5); title('Pho pha x(t)'); grid on;</pre>

Kết quả

Câu a:



Câu b:

